SASIDHAR GUTHI

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years designing and operating large-scale, AWS-native data platforms across healthcare, retail, and financial domains. Deep expertise in ETL/ELT pipeline design using AWS Glue, Lambda, Step Functions, Athena, Kinesis, and Apache Airflow — with a strong foundation in data modeling (Kimball dimensional, star/snowflake schemas, conformed dimensions) and OLAP systems including Redshift and Snowflake. Proven track record migrating legacy batch pipelines to modern lakehouse architectures, enforcing data quality and lineage standards, and collaborating with BI engineers, data scientists, and business stakeholders to ship scalable, reliable data solutions. Cleared AWS Glue bottlenecks reducing job runtime by 35%, cut analytics latency by 40%, and shipped a production event validation platform serving telemetry at Walmart/Vizio.

CORE COMPETENCIES

ETL/ELT & Pipeline Design: AWS Glue, Lambda, Step Functions, Athena, Apache Airflow, Apache Spark, EMR, PySpark, Scala

Data Modeling & OLAP: AWS Redshift, Snowflake, BigQuery, Oracle, Dimensional Modeling (Kimball), Star & Snowflake Schemas, Conformed Dimensions, Entity Models

Lakehouse & Modernization: Apache Iceberg, AWS Lake Formation, Redshift Spectrum, Parquet/ORC optimization, S3-based data lakes

Data Quality & Governance: Schema enforcement, data lineage, freshness monitoring, PII/HIPAA compliance, data drift detection, naming conventions, validation-first design

Secured Data Access: AWS IAM roles & policies, VPC endpoint routing, fine-grained access control, HIPAA-compliant data platforms

Non-Relational & Distributed Stores: DynamoDB (key-value), S3 (object storage), OpenSearch, PostgreSQL, Aurora RDS, SQL Server

Streaming & Real-Time: Amazon Kinesis (Streams, Firehose, Analytics), Kafka, SQS, SNS

CI/CD & Operational Excellence: GitHub Actions, GitLab CI, AWS CodeBuild, Docker, Kubernetes, Terraform, on-call & incident response

AI & LLM Data Engineering: RAG pipeline design, vector embeddings, LLM integration, prompt engineering, vector stores, document chunking & ingestion, OpenAI APIs, LangChain

Programming & Querying: Python, SQL, PySpark, Scala, Bash

PROFESSIONAL EXPERIENCE

Data Engineer | Walmart **| Vizio** | Addison, TX *Apr 2024 – Present*

Built and owned end-to-end data infrastructure for the Vizio Mobile App platform — spanning telemetry ingestion, streaming analytics, ML feature pipelines, and a full-stack event validation system.

- Reduced streaming analytics latency by 40% by designing and implementing a real-time ETL pipeline using Kinesis Firehose + EMR Spark + Airflow-orchestrated Snowflake loads, replacing a legacy hourly batch process.
- Cut AWS Glue job runtime by 35% and lowered compute costs by identifying pipeline bottlenecks, right-sizing worker types, and tuning partition strategies — directly resolving technical debt across 12 production ETL jobs.
- Enforced end-to-end data quality and lineage by shipping a production event validation platform (Lambda + API Gateway + FastAPI) that applied JSON schema contracts across Kafka and Kinesis streams, eliminating silent data failures in telemetry.
- Migrated legacy financial batch workflows to parallelized Python + Boto3 pipelines, reducing processing time by 50% through concurrent S3 reads and distributed AWS integration.
- Designed and maintained conformed dimension models and entity models in Snowflake to support downstream BI consumption, standardizing naming conventions and modeling patterns across the analytics org.

- Built secured access controls for large telemetry datasets using AWS IAM roles, S3 bucket policies, and API Gateway authorizers — ensuring only validated consumers could access raw and curated data layers.
- Established CI/CD pipelines in GitLab with automated unit/integration testing and deployment gates, reducing pipeline release cycle from weekly to daily and enabling confident data model iteration.
- Led load testing and latency profiling with JMeter and Locust, identifying and resolving a p99 bottleneck degrading downstream Snowflake write performance under peak ingestion load.
- Designed and deployed a RAG (Retrieval-Augmented Generation) pipeline integrating vector-based similarity search with an LLM layer to enable dynamic SQL generation, reducing irrelevant model calls by ~30% and enabling self-service analytics for non-technical stakeholders.

Data Engineer | Merck **Pharmaceuticals** | Charlotte, NC *Jun 2023 – Apr 2024*

Led data engineering for HIPAA-compliant EHR and clinical data platforms, building scalable ingestion and real-time processing pipelines that powered clinical reporting and claims analytics.

- Reduced clinical reporting pipeline SLA from 6 hours to under 90 minutes by migrating legacy Oracle + AWS Data Pipeline workflows to optimized AWS Glue with PySpark, eliminating sequential processing bottlenecks.
- Processed 10M+ clinical records daily using Apache Spark on EMR across Redshift OLAP workloads, enabling claims analysis that previously required manual extraction from siloed Oracle databases.
- Designed conformed dimension models and entity models in Redshift to standardize clinical data definitions across 5 downstream BI and regulatory reporting teams, enforcing naming conventions and data modeling best practices.
- Achieved HIPAA-compliant secured data access at sub-50ms latency by integrating AWS Lambda with Aurora RDS using fine-grained IAM role policies and VPC endpoint routing — replacing an insecure legacy polling mechanism.
- Built real-time EHR data feeds with Kinesis Data Streams + Kinesis Analytics, delivering 15-second end-to-end freshness SLA for clinical dashboards and supporting incident response for pipeline anomalies.
- Implemented data quality framework with schema validation, lineage tracking, and alerting across 20+ MWAA Airflow DAGs, reducing bad-data incidents reaching downstream BI by 80%.
- Authored complex analytical SQL against Oracle source systems — joins across 15+ clinical schemas, stored procedures, and incremental extraction queries — serving as the primary data engineer owning Oracle-to-Redshift ingestion for claims and EHR reporting pipelines.
- Reduced time-to-insight for clinical trial reporting from 3 days to 4 hours by redesigning the Oracle extraction layer with incremental CDC logic, eliminating full-table scans and cutting pipeline runtime by 65% — directly unblocking a regulatory submission deadline.

Data Engineer | Infosys | Hyderabad, India *Sep 2019 – Aug 2021*

Built event-driven and serverless data pipelines for high-volume financial trading systems, delivering real-time fraud detection and capital markets analytics.

- Cut pipeline storage costs by 40% by migrating data serialization from CSV to columnar Parquet and ORC formats across 5 high-volume trading data pipelines, improving downstream Spark read throughput by 3×.
- Enabled real-time fraud detection analytics by building Spark Streaming pipelines that processed 500K+ trade events/hour with sub-second latency using Lambda + SQS + Step Functions orchestration.
- Reduced manual operational toil by 70% through serverless deployment automation using S3 event triggers, Lambda, and CloudWatch alarms — replacing manual script execution for batch ingestion.
- Designed asynchronous messaging architecture (Lambda + SQS + Step Functions) to decouple ingestion from transformation, achieving fault-tolerant pipeline execution with automatic retries and dead-letter queue monitoring.

EDUCATION

M.S., Data Science — University of Maryland, Baltimore County *2021 – 2023*

B.Tech., Electrical & Electronics Engineering — Andhra University *2015 – 2019*